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Insulated earthing switch for an air insulated or mixed switchgear

Abstract

The application concerns an air insulated or mixed air and gas insulated switchgear assembly including a switchgear and an earthing switch. The switchgear includes a switchgear housing. The earthing switch includes a switch housing. The earthing switch includes an insulator flange including a metallic body and a dielectric insulating layer on the metallic body. The switch housing is mechanically connected to the switchgear housing through the insulator flange.

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Background/Summary

TECHNICAL FIELD

(1) The invention belongs to the field of medium to high-voltage technology, i.e., 52 kV or higher. In particular, the invention relates to electrical insulation for air-insulated switchgear assemblies (AIS) or mixed air and gas insulated switchgear assemblies.

BACKGROUND

(2) FIG. 1 shows an existing gas insulated switchgear assembly 5. Such a gas insulated assembly has to withstand higher voltage than an air-insulated switchgear assembly or a mixed switchgear assembly.

(3) The gas insulated switchgear assembly 5 includes an insulating earthing switch 15 and a gas insulated switchgear 2. The earthing switch 15 is separated from the housing 3 of the gas insulated

switchgear **2** by an insulator **17** which is made of plastic material such as an epoxy resin.

(4) During normal operation, the earthing switch **5** is electrically connected to the gas insulated switchgear **2** by a ground connection **19**. The ground connection **19** is a short circuit bridge between an earthing switch housing **13** and a switchgear housing **3**.

(5) For measurements in which a measured signal is applied to or tapped off over the contact system of the insulating earthing switch **15**, the ground connection **19** is removed.

(6) Such an existing insulated earthing switch has a limited resilience to outdoor application. In particular, the insulator of the earthing switch has limited mechanical resistance and it may break. Besides, the earthing switch has limited environmental durability because of the plastic material insulator.

SUMMARY

(7) The invention relates to an air insulated or mixed air and gas insulated switchgear assembly. The switchgear assembly includes an air insulated or mixed air and gas insulated switchgear and an earthing switch. The switchgear includes a switchgear housing and a central conductor extending within the switchgear housing. The earthing switch includes a switch housing, a movable contact rod and a fixed electrical contact. The fixed electrical contact is secured to the central conductor. The movable contact rod is movable relative to the switch housing and to the fixed electrical contact.

(8) The movable contact rod electrically engages the fixed electrical contact when the earthing switch is in a closed position. The movable contact rod is electrically disconnected from the fixed electrical contact when the earthing switch is in an opened position.

(9) According to the invention, the earthing switch includes an insulator flange comprising a metallic body and a dielectric insulating layer on the metallic body, and the switch housing is mechanically connected to the switchgear housing through the insulator flange.

(10) Thanks to the insulator flange of the gas insulated switchgear assembly of the invention, the earthing switch has a higher resilience to outdoor applications. In particular, the insulator flange has high mechanical resistance and it will not break easily. The insulator flange may be easily recycled. The insulated earthing switch has higher environmental durability.

(11) Preferred features of the invention are recited in the dependent claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) This invention will be better understood after reading the following description of example embodiments that are in no way limitative, wherein:

(2) FIG. 1 schematically and partially shows a known gas insulated switchgear assembly in longitudinal cross section;

(3) FIG. 2 schematically shows a portion of a gas insulated switchgear assembly according to a first embodiment of the invention in a perspective view;

(4) FIG. 3 schematically shows a portion of the gas insulated switchgear assembly according to a first embodiment of the invention in longitudinal cross section;

(5) FIG. 4 schematically shows a portion of the connection between a switchgear and an earthing switch of the switchgear assembly according to the first embodiment and in cross section;

(6) FIG. 5a is a schematic circuit diagram of the gas insulated switchgear assembly of the first embodiment when the earthing switch is in a closed position;

(7) FIG. 5b is a schematic circuit diagram of the gas insulated switchgear assembly of the first embodiment in an opened position during normal operation of the switchgear assembly.

DETAILED DESCRIPTION

(8) FIGS. 2 to 4 show an air insulated or mixed air and gas insulated switchgear assembly **1**. The

gas is an arc extinguishing gas such as SF₆ or such as a mixture of fluoronitrile, fluoroketones, carbon dioxide and/or dioxygen.

(9) The switchgear assembly **1** includes an air insulated or mixed air and gas insulated switchgear **2** for each phase of an AC electric line, an earthing switch **10** for each phase of the AC electric line, a ground connection **9** and fasteners **20**.

(10) Each switchgear **2** includes a switchgear housing **3**, a central conductor **7** and insulator walls **6**.

(11) FIG. **3** shows more specifically a switchgear **2** and an earthing switch **10** of one of the phases of the AC electric line.

(12) With reference to FIG. **3**, the switchgear housing **3** encloses a gas chamber **2a** which is filled by air or a mix of air and gas. The switchgear housing **3** includes a mounting flange **8** to which the earthing switch **10** is fixed. The switchgear housing **3** surrounds the central conductor **7**. The switchgear housing **3** is connected to the ground independently from the earthing switch **10**.

(13) The central conductor **7** includes electrically active parts of the switchgear **2**. The central conductor **7** extends longitudinally along the longitudinal direction X-X of the switchgear.

(14) The insulator walls **6** each hold the central conductor **2** relative to the switchgear housing **3**. The insulator walls **6** each extend radially relative to the longitudinal direction X-X of the switchgear between the central conductor **7** and the switchgear housing **3**.

(15) With reference to FIG. **2** and FIG. **3**, each earthing switch **10** includes a switch housing **13**, a movable contact rod **12**, a drive mechanism **11**, a fixed electrical contact **14** and an insulator flange **16**. Each earthing switch **10** is configured to ground the central conductor **7** of the switchgear **2** of the corresponding phase.

(16) In the disclosed embodiment, the longitudinal direction Y-Y of the earthing switch is substantially perpendicular to the longitudinal direction X-X of the switchgear of the corresponding phase.

(17) With reference to FIG. **3**, the switch housing **13** encloses at least part of the movable contact rod **12** and at least part of the drive mechanism **11**. The switch housing **13** is fixed to the switchgear housing **13** by the insulator flange **16** and by the fasteners **20**.

(18) The fixed electrical contact **14** is fixed to the central conductor **7**. The fixed electrical contact **14** is for example a tulip type female conductor. In particular, the fixed electrical contact **14** may comprise contact fingers for grasping the movable contact rod **14**.

(19) The movable contact rod **12** is annular along the longitudinal direction Y-Y of the earthing switch.

(20) The movable contact rod **12** is movable along the longitudinal direction Y-Y of the earthing switch relative to the switch housing **13** and to the fixed electrical contact **14** between an elongated position and a retracted position.

(21) The movable contact rod **12** in the elongated position electrically and mechanically engages the fixed electrical contact **14**. When the movable contact rod **12** electrically engages the fixed electrical contact **14**, the earthing switch **10** is in a closed position.

(22) The movable contact rod **12** in the retracted position is electrically disconnected from the fixed electrical contact. When the movable contact rod **12** is electrically disconnected from the fixed electrical contact **14**, the earthing switch **10** is in an opened position.

(23) The drive mechanism **11** is configured to move the movable contact rod **12** relative to the switch housing **13** and to the fixed electrical contact **14** between the elongated position and the retracted position with a drive movement **11a** along the longitudinal direction Y-Y of the earthing switch.

(24) The insulator flange **16** comprises a metallic body **30** and a dielectric insulating layer **31**. The switch housing **13** is mechanically connected to the switchgear housing **3** through the insulator flange **16**. The insulator flange **16** electrically insulates the switchgear housing **3** from the switch housing **13**.

(25) The electrically insulating layer **31** covers entirely the metallic body **30** in the disclosed embodiment. The electrically insulating layer **31** forms the entire external surface of the insulator flange **16** in the disclosed embodiment.

(26) The insulating layer **31** includes an oxidation layer. In the first disclosed embodiment, the oxidation layer is made from the same metallic material as the metallic material of the metallic body **30** of the insulator flange **16**.

(27) The oxidation layer is made by anodic oxidation of the metallic body **30** of the insulator flange **16**. The metallic body **30** is for example made of aluminum and the oxidation layer is for example made by hard anodization of the metallic body **30**. The oxidation layer is in particular made of alumina in that case.

(28) The fasteners **20** include first fasteners **21, 23, 27** and second fasteners **22, 25, 29**. The fasteners **20** fasten the switch housing **13** to the switchgear housing **3** through the insulator flange **16**.

(29) The first fasteners **21, 23, 27** are arranged in a first circular row for fastening the switch housing **13** to the insulator flange **16**. The first fasteners each include a metallic screw **21**, a metallic washer **23** and a metallic peg **29** in the disclosed embodiment.

(30) The second fasteners **22, 25, 29** are arranged in a second circular row concentric with the first circular row and radially spaced from the first circular row. The second fasteners **22, 25, 29** fasten the insulator flange **16** to the switchgear housing **3**.

(31) The second fasteners **22, 23, 25** are insulated fasteners **22, 25, 29**. The second fasteners each include a metallic screw **22**, a dielectric insulating washer **25** and a dielectric insulating peg **29**.

(32) The dielectric insulating washer **25** are each made of a material containing plastic. The dielectric insulating peg **29** are also made of a material containing plastic. The dielectric insulating washer **25** and dielectric insulating peg **29** are configured to limit wear of the insulating layer **31**. They also contribute to electrically insulating the switchgear housing **3** from the switch housing **13**.

(33) The ground connection includes an earthing frame **9**. The ground connection **9** is configured to electrically insulate the switchgear housing **3** from the switch housing **13** together with the insulator flange **16**.

(34) The ground connection **9** is configured to electrically connect the switch housing **13** to the ground, when the earthing switch **10** is in the opened position and when the earthing switch **10** is in the closed position. The ground connection **9** is electrically disconnected from the switchgear housing **3**.

(35) The earthing frame **9** is secured to the earthing switch **10** housing **13** of each phase. The earthing frame **9** includes earthing rods **90** and an earth connection **91**.

(36) The earthing rods **90** each extend between two earthing switch housings **13** of two different phases of the switchgear assembly **1**. The earth connection **91** connects the earthing rods **90** to the ground independently from each switchgear housing **3**.

(37) With reference to FIG. **5a**, the ground connection **9** is electrically connected to the central conductor **7**, to the movable contact rod **12** and to the switch housing **13**, when the earthing switch **10** is in the closed position.

(38) During grounding operation of the switchgear assembly **1**, each earthing switch **10** is closed and each movable rod **12** is in the elongated position.

(39) With reference to FIG. **5b**, the ground connection **9** is electrically connected to the movable contact rod **12** and to the switch housing **13**, when the earthing switch **10** is in the opened position.

(40) During normal operation of the switchgear assembly **1**, each earthing switch **10** is opened and each movable rod **12** is in the retracted position. The central conductor **7** of each switchgear **2** is under medium to high voltage and/or current.

(41) Thanks to each insulator flanges **16** of the gas insulated switchgear assembly, each earthing switch **10** has a higher resilience to outdoor applications. In particular, the insulator flange **16** has high mechanical resistance and it will not break easily. The insulator flange **16** may be easily

recycled. The earthing switch **10** has thus higher environmental durability.

(42) The above described embodiments may of course be modified by the man of ordinary skill in the art.

(43) The insulator flange **16** may be deprived of electrically insulating layer **31** near the fasteners **20**. At the very least, the electrically insulating layer **31** covers the metallic body **30** at least on a first contact surface of the insulator flange **16** with the switchgear housing **3**. In addition or alternatively, the electrically insulating layer **31** covers the metallic body **30** at least on a second contact surface of the insulator flange **16** with the switch housing **13**.

(44) The insulator layer may include a dielectric coating, instead of and in addition to the oxidation layer.

(45) The oxidation layer may be made of another metallic material from the metallic material of the metallic body **30** of the insulator flange **16**, for example when the oxidation layer is deposited or cladded onto the metallic body **30**.

(46) The fasteners **20** may include a rivet, a pin and/or a nut and in addition to or instead of screws and pegs.

(47) The first fasteners **21**, **23**, **27** may be insulated fasteners and/or the second fasteners **22**, **23**, **25** may be non-insulated metallic fasteners.

(48) It should be apparent that the foregoing relates only to certain embodiments of this application and resultant patent. Numerous changes and modifications may be made herein by one of ordinary skill in the art without departing from the general spirit and scope of the invention as defined by the following claims and the equivalents thereof.

Claims

1. An air insulated or mixed air and gas insulated switchgear assembly, comprising: an air insulated or mixed air and gas insulated switchgear comprising a switchgear housing and a central conductor extending within the switchgear housing; an earthing switch comprising a switch housing, a movable contact rod and a fixed electrical contact; wherein the fixed electrical contact is secured to the central conductor; wherein the movable contact rod is movable relative to the switch housing and to the fixed electrical contact; wherein the movable contact rod electrically engages the fixed electrical contact when the earthing switch is in a closed position, and wherein the movable contact rod is electrically disconnected from the fixed electrical contact when the earthing switch is an opened position; wherein the earthing switch comprises an insulator flange comprising a metallic body and a dielectric insulating layer on the metallic body, and in that the switch housing is mechanically connected to the switchgear housing through the insulator flange.

2. The switchgear assembly according to claim 1, wherein the dielectric insulating layer forms substantially an entire external surface of the insulator flange.

3. The switchgear assembly according to claim 1, wherein the dielectric insulating layer comprises an oxidation layer, wherein the oxidation layer is preferably made from a same metallic material as a metallic material of the metallic body.

4. The switchgear assembly according to claim 3, wherein the oxidation layer is made by anodic oxidation of the metallic body of the insulator flange.

5. The switchgear assembly according to claim 3, wherein the oxidation layer is made by hard anodization of the metallic body.

6. The switchgear assembly according to claim 1, wherein the switchgear assembly comprises fasteners for fastening the switch housing to the switchgear housing through the insulator flange.

7. The switchgear assembly according to claim 6, wherein the fasteners include first fasteners arranged in a first annular row for fastening the switch housing to the insulator flange, and wherein the fasteners include second fasteners arranged in a second annular row concentric with the first annular row and radially spaced from the first annular row, wherein the second fasteners fasten the

insulator flange to the switchgear housing.

8. The switchgear assembly according to claim 6, wherein the fasteners include insulated fasteners each including a metallic screw and a dielectric insulating washer made of a material containing plastic.

9. The switchgear assembly according to claim 6, wherein the insulated fasteners each further include a dielectric insulating peg made of a material containing plastic.

10. The switchgear assembly according to claim 1, wherein the switchgear assembly comprises a ground connection for electrically connecting the switch housing to a ground, when the earthing switch is in the opened position and when the earthing switch is in the closed position.

11. The switchgear assembly according to claim 10, wherein the ground connection is electrically connected the central conductor, to the movable contact rod and to the switch housing when the earthing switch is in the closed position, wherein the ground connection is electrically disconnected from the switchgear housing when the earthing switch is in the opened position.

12. The switchgear assembly according to claim 10, wherein the switchgear assembly comprises an air insulated or mixed air and gas insulated switchgear for each phase of an AC electric line; wherein the switchgear assembly comprises an earthing switch for each phase of an AC electric line; wherein each switchgear includes a switchgear housing and a central conductor extending inside the switchgear housing; wherein each earthing switch comprising a switch housing, a movable contact rod, a fixed electrical contact, and an insulator flange comprising a metallic body and a dielectric insulating layer on the metallic body; wherein the ground connection comprises an earthing frame secured to the earthing switch housing of each phase, wherein the earthing frame comprises earthing rods each extending between two earthing switch housings of the switchgear assembly.
